

Diabetes Disease Classification Dataset

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1. Overview of the Dataset

Name of the dataset	: Diabetes Disease Classification Dataset
File name	: Diabetes_MCDD.csv
File type	: Comma Separated Value (.csv)
No. of columns	: 19
No. of rows	: 2,000+
No. of instances	: ~800–1,000 unique patients
No. of attributes	: 19 (17 clinical/laboratory + 2 target)
No. of class labels	: 3 (Diabetes_Status: Non-Diabetic, Pre-Diabetic, Diabetic)
Source	: Sardar Ranjit Singh Memorial Hospital (SRS) and Research Center, Ambikapur, Chhattisgarh, India
Creator	: Deepshikha Gupta, Research Scholar, NIT Silchar, Assam, India

2. Descriptions of the Attributes

Attribute	: Description
Age	: Age of the patient (years)
Sex	: Gender (0 = Female, 1 = Male)
FBS_mg_dL	: Fasting Blood Sugar (mg/dL) – measured after overnight fast
PPBS_mg_dL	: Post-Prandial Blood Sugar (mg/dL) – 2 hours after meal
HbA1c_percent	: Glycated Haemoglobin (%) – average blood glucose over 2-3 months
BMI	: Body Mass Index (kg/m ²), calculated from weight and height
Weight_kg	: Weight of the patient (kg)
Height_cm	: Height of the patient (cm)
BP_Systolic_m mHg	: Systolic Blood Pressure (mmHg)
Pulse_bpm	: Pulse rate (beats per minute)
Total_Cholesterol_mg_dL	: Total Cholesterol (mg/dL)
HDL_Cholesterol_mg_dL	: High-Density Lipoprotein (HDL) Cholesterol (mg/dL)

LDL_Cholesterol_mg_dL	: Low-Density Lipoprotein (LDL) Cholesterol (mg/dL)
Triglycerides_mg_dL	: Triglycerides level (mg/dL)
Serum_Creatinine_mg_dL	: Serum Creatinine (mg/dL) – indicator of kidney function
TSH_uIU_mL	: Thyroid-Stimulating Hormone (uIU/mL) – thyroid function
Diagnosis_Temp	: Free-text field listing additional diagnoses/comorbidities (e.g., hypertension, hypothyroidism)
Diabetes_Status	: Final clinical diabetes classification (Non-Diabetic, Pre-Diabetic, Diabetic)
Diabetes_Target	: Numeric encoding of Diabetes_Status (0 = Non-Diabetic, 1 = Pre-Diabetic, 2 = Diabetic)

3. Description of Class Labels

Class label name	: Diabetes_Status (or Diabetes_Target for numeric modelling)
0	: Non-Diabetic (No diabetes diagnosis)
1	: Pre-Diabetic (Elevated glucose/HbA1c, but not reaching diabetic thresholds)
2	: Diabetic (Diagnosed diabetes mellitus)

4. Sample Image

Age	Sex	FBS_mg_dL	PPBS_mg_dL	HbA1c_percent	BMI	Weight_kg	Height_cm	BP_Systolic_mmHg	Pulse_bpm	Total_Cholesterol_mg_dL	HDL_Cholesterol_mg_dL	LDL_Cholesterol_mg_dL	Triglycerides_mg_dL	Serum_Creatinine_mg_dL	TSH_uIU_mL	Diagnosis_Temp	Diabetes_Status	Diabetes_Target
31	0			99.57	16.79363	38.8	152	70	88						1.351		Diabetic	2
24	0				17.18317	39.7	152	70	88								Non-Diabetic	0
53	1			14.6	20.71006	50.4	156	129	88								Diabetic	2
25	1				24.42742	65.7	164	118	88						2.87	HYPOTHYROID	Non-Diabetic	0
25	1				24.42742	65.7	164	118	88						2.87		Non-Diabetic	0
25	1				24.42742	65.7	164	118	88						2.87		Non-Diabetic	0
46	1	117		10.3	24.65279	65.5	163	110	92		6.35	23.4	117	73.62			Diabetic	2
53	0	149	195	16.1	24.49823	58.1	154	120						112.34	1.772		Diabetic	2
31	0				31.56603	84.9	164	130	88						0.688		Non-Diabetic	0
70	0		148		35.67958	77.1	147	91	88	192	4.57	120	150				Non-Diabetic	0
70	0		148		35.67958	77.1	147	91	88	192	4.57	120	150			HYPERTENSION	Diabetic	2
70	1	113	169		22.24175	65.8	172	86	88								Pre-Diabetic	1
70	1	113	169		22.24175	65.8	172	86	88								Pre-Diabetic	1
58	0	155.2	240.28		30.54283	66	147	70		192.68	4.89	35.13	175.65	104.35	1.76		Diabetic	2
58	0	155.2	240.28		30.54283	66	147	70		192.68	4.89	35.13	175.65	104.35	1.76	DIABETES MELLI	Diabetic	2
58	0	155.2	240.28		30.54283	66	147	70		192.68	4.89	35.13	175.65	104.35	1.76	DIABETES MELLI	Diabetic	2
58	0	155.2	240.28		30.54283	66	147	70		192.68	4.89	35.13	175.65	104.35	1.76		Diabetic	2
65	1	145	102	125	30.91485	85.6	166.4	151	97	131	50	73.8	166	93	2.61		Diabetic	2
65	1	145	102	125	30.91485	85.6	166.4	151	97	131	50	73.8	166	93	2.61	DIABETES MELLI	Diabetic	2
65	1	145	102	125	30.91485	85.6	166.4	151	97	131	50	73.8	166	93	2.61		Diabetic	2
65	1	145	102	125	30.91485	85.6	166.4	151	97	131	50	73.8	166	93	2.61		Diabetic	2
65	1	145	102	125	30.91485	85.6	166.4	151	97	131	50	73.8	166	93	2.61		Diabetic	2
48	0	123	178			65.7											Pre-Diabetic	1
50	1	145	262		22.46315	63.4	168	70	85	246	57	18.8	94				Diabetic	2

5. Primary Research Usability

This dataset, despite its data quality challenges, holds substantial potential for a range of clinical and epidemiological research investigations. After appropriate cleaning, deduplication, and outlier treatment, the records can be used to address the following research questions:

- Diabetes risk prediction and stratification

The availability of multiple glycemic indices (FBS, PPBS, HbA1c), anthropometric measures (BMI, weight, height), cardiovascular risk factors (blood pressure, lipid profile), and thyroid function data allows the construction and validation of machine learning models to predict diabetes status (Non-Diabetic, Pre-Diabetic, Diabetic).

- Identification of key clinical predictors

Through feature importance analysis (e.g., using random forests or logistic regression), the dataset can reveal which variables most strongly discriminate between diabetic and non-diabetic.

- Comorbidity pattern analysis
The free-text field `diagnosis_temp` contains multiple co-occurring conditions (hypertension, hypothyroidism, dyslipidemia, ischemic heart disease, etc.). After parsing and standardizing these diagnoses, the dataset can be used to study comorbidity clusters and their association with glycemic control and diabetes severity.
- Missing data mechanism studies
The extensive missingness in laboratory values offers an opportunity to investigate patterns of missing data (missing completely at random, missing at random, or missing not at random) and to compare the performance of different imputation techniques (mean/median imputation, multiple imputation, or model-based methods) in a clinical prediction context.
- Development of early-warning systems
The dataset includes Pre-Diabetic cases, which are a high-priority target for intervention. Machine learning classifiers trained on this data could be deployed in low-resource settings to flag high-risk individuals for further testing and lifestyle modification.